```
FUNCTION YEHSM_ECEL_DATA_SAVE.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     REFERENCE(ES_AUDIT_HEADER) TYPE  YSROOT1_D
*"     REFERENCE(EV_TOT_OBS_NO) TYPE  YY_OBS_NO
*"     REFERENCE(EV_COORDINATOR_ROLE) TYPE  FLAG
*"     REFERENCE(EV_IMPLEMENTOR_ROLE) TYPE  FLAG
*"     REFERENCE(EV_CWC_ROLE) TYPE  FLAG
*"     REFERENCE(EV_VIEW_ROLE) TYPE  FLAG
*"     REFERENCE(MO_SVC_MNGR) TYPE REF TO  /BOBF/IF_TRA_SERVICE_MANAGER
*"  TABLES
*"      IT_AUDITORS TYPE  YTTAUDITORS_D
*"      IT_EXCEL_DATA TYPE  YTTEHS_EXCEL_D
*"----------------------------------------------------------------------
TYPES: BEGIN OF ty_lines,
          line(132) TYPE c,
        END OF ty_lines,

       BEGIN OF ty_text,
          obs_no  TYPE yy_obs_no,
          type(5) TYPE c,
          text    TYPE string,
        END OF ty_text,

        BEGIN OF ty_message,
          Row TYPE int8,
          Message TYPE string,
          END OF ty_message.

  DATA : lt_status           TYPE STANDARD TABLE OF dd07v,
         ls_status           TYPE dd07v,
         ls_audit_obs        TYPE ysaudit_observations_d,
         lv_rcdate           TYPE datum,
         lv_aufrom           TYPE datum,
         lv_tgt_date         TYPE datum,
         lv_delayed_text     TYPE string,
         lt_delay_reason     TYPE STANDARD TABLE OF ty_lines,
         ls_delay_reason     TYPE ty_lines,
         lv_range_nr         TYPE inri-nrrangenr,
         lv_object           TYPE inri-object,
         lv_audit_number(10) TYPE n,
         lt_text             TYPE TABLE OF ty_text,
         ls_text             TYPE ty_text,
         lt_return           TYPE TABLE OF ty_message,
         ls_return           TYPE ty_message,
         lv_row              TYPE int8 VALUE IS INITIAL,
         ls_audit_header     TYPE YSROOT1_D,
         lv_save_mode(1)     TYPE c VALUE 'S'.

*-- GET NEW NUMBER FOR AUDIT
  DATA : lt_mod              TYPE /bobf/t_frw_modification,
         lr_audit            TYPE REF TO ysroot1,
         lr_auditors         TYPE REF TO ysauditors,
         lr_obs              TYPE REF TO ysaudit_observations,
         lr_aud_obs          TYPE REF TO ysaudit_obs_text,
         lo_change           TYPE REF TO /bobf/if_tra_change,
         lo_message          TYPE REF TO /bobf/if_frw_message,
         lt_messages         TYPE /bobf/t_frw_message_k,
         lt_audit_chang      TYPE STANDARD TABLE OF yehs_audit_chang,
         ls_audit_chang      TYPE  yehs_audit_chang,
```

```
         mo_txn_mngr         TYPE REF TO /bobf/if_tra_transaction_mgr,
*         mo_svc_mngr         TYPE REF TO /bobf/if_tra_service_manager,
         lv_rejected         TYPE boole_d,
         lx_bopf_ex          TYPE REF TO /bobf/cx_frw,
         lv_err_msg          TYPE string.
*         lr_activity         TYPE REF TO ysactivit_log,
*         gs_audit_key           TYPE ydehs_audit.

*DATA:  gs_milestone_final     TYPE ydmilestones, "ysmilestones_d,.
*       gt_audit_key           TYPE STANDARD TABLE OF ydehs_audit,
*       lv_flag TYPE flag.

FIELD-SYMBOLS : <ls_message>      LIKE LINE OF lt_messages,
                <fs_auditors> TYPE ysauditors_d,
                <ls_mod>      LIKE LINE OF lt_mod.

ls_audit_header = es_audit_header.

ls_audit_obs-status = 'O'.
ls_audit_obs-obs_no = 00001.

lv_range_nr = '01'.
lv_object = 'YAUDIT'.

*—— Get Reference of BOBF Transaction Manager
*  IF mo_svc_mngr IS NOT BOUND.
    mo_txn_mngr = /bobf/cl_tra_trans_mgr_factory=>get_transaction_manager( ).
*  ENDIF.
*********************************************Authorization Code
  IF lt_status[] IS INITIAL.
    CALL FUNCTION 'GET_DOMAIN_VALUES'
      EXPORTING
        domname    = 'YY_ENH_STATUS'
      TABLES
        values_tab = lt_status.
  ENDIF.

  READ TABLE lt_status INTO ls_status WITH KEY domvalue_l = ls_audit_obs-status.
  IF sy-subrc = 0.
  IF ev_cwc_role = 'X'.
    AUTHORITY-CHECK OBJECT 'YAUDIT_CWC'
    ID 'WERKS' FIELD ls_audit_header-audit_plant
    ID 'YAUD_STAT' FIELD ls_audit_obs-status.
    IF sy-subrc NE 0.
      MESSAGE i030(ye01) WITH ls_status-ddtext ls_audit_header-audit_plant DISPLAY LIKE 'E'.
*      p_gv_flag = 'X'.
      EXIT.
    ENDIF.

  ELSEIF ev_implementor_role = 'X' AND ev_coordinator_role IS INITIAL.
    AUTHORITY-CHECK OBJECT 'YAUDIT_IM'
    ID 'WERKS' FIELD ls_audit_header-audit_plant
    ID 'YAUD_STAT' FIELD ls_audit_obs-status.
    IF sy-subrc NE 0.
      MESSAGE i030(ye01) WITH ls_status-ddtext ls_audit_header-audit_plant DISPLAY LIKE 'E'.
*      p_gv_flag = 'X'.
      EXIT.
```

```
119      ENDIF.
120
121    ELSEIF ev_implementor_role = 'X' AND ev_coordinator_role = 'X'.
122      AUTHORITY-CHECK OBJECT 'YAUDIT_IM'
123      ID 'WERKS' FIELD ls_audit_header-audit_plant
124      ID 'YAUD_STAT' FIELD ls_audit_obs-status.
125      IF sy-subrc NE 0.
126        AUTHORITY-CHECK OBJECT 'YAUDIT_CO'
127        ID 'WERKS' FIELD ls_audit_header-audit_plant
128        ID 'YAUD_STAT' FIELD ls_audit_obs-status.
129        IF sy-subrc NE 0.
130          MESSAGE i030(ye01) WITH ls_status-ddtext ls_audit_header-audit_p
     lant DISPLAY LIKE 'E'.
131  *         p_gv_flag = 'X'.
132          EXIT.
133        ENDIF.
134      ENDIF.
135
136    ELSEIF ev_coordinator_role = 'X'.
137      AUTHORITY-CHECK OBJECT 'YAUDIT_CO'
138      ID 'WERKS' FIELD ls_audit_header-audit_plant
139      ID 'YAUD_STAT' FIELD ls_audit_obs-status.
140      IF sy-subrc NE 0.
141        MESSAGE i030(ye01) WITH ls_status-ddtext ls_audit_header-audit_pla
     nt DISPLAY LIKE 'E'.
142  *       p_gv_flag = 'X'.
143        EXIT.
144      ENDIF.
145    ENDIF.
146    ELSE.
147      EXIT.
148      ENDIF.
149  ***************************************End of Authorization Code**********
     ***********************************
150  DATA: lv_date TYPE datum.
151  LOOP at it_excel_data INTO DATA(ls_audit_data).
152
153  *ls_audit_obs-status = 'O'.
154  *ls_audit_obs-obs_no = 00001.
155  *—— If Audit was not entered in system for more than 21 days, Ask user t
     o input reason of delay
156
157  CLEAR: lv_rcdate, lv_aufrom, lv_tgt_date, lv_delayed_text, lv_audit_numb
     er.
158  *CLEAR: lt_delay_reason[], ls_delay_reason.
159
160  PERFORM date_conv using ls_audit_data-rcdate CHANGING lv_date.
161  lv_rcdate = lv_date.
162  CLEAR lv_date.
163  *  CALL FUNCTION 'CONVERT_DATE_INPUT'
164  *  EXPORTING
165  *    input  = ls_audit_data-rcdate
166  *  IMPORTING
167  *    output = lv_rcdate
168  *  EXCEPTIONS
169  *    OTHERS = 1.
170  PERFORM date_conv using ls_audit_data-aufrom CHANGING lv_date.
171  lv_aufrom = lv_date.
172  CLEAR lv_date.
173  *    CALL FUNCTION 'CONVERT_DATE_INPUT'
174  *  EXPORTING
```

```
*    input  = ls_audit_data-aufrom
*  IMPORTING
*    output = lv_aufrom
*  EXCEPTIONS
*    OTHERS = 1.

    CALL FUNCTION 'CONVERT_DATE_INPUT'
  EXPORTING
    input  = ls_audit_data-tgtdate
  IMPORTING
    output = lv_tgt_date
  EXCEPTIONS
    OTHERS = 1.

*  IF ( sy-datum - lv_rcdate ) GT 21. "gs_audit_header-audit_rec_date ) GT 21.
*    DO.
*      CALL FUNCTION 'TERM_CONTROL_EDIT'
*        EXPORTING
*          titel          = 'Enter Reason For Delayed Entry In SAP'
*          langu          = 'E'
*        TABLES
*          textlines      = lt_delay_reason
*        EXCEPTIONS
*          user_cancelled = 1.
*
*      IF lt_delay_reason[] IS INITIAL.
*        MESSAGE 'Please enter reason for delayed entry in SAP for Audit/Activity.' TYPE 'I'.
*      ELSE.
*        EXIT.
*      ENDIF.
*    ENDDO.
*
*    LOOP AT lt_delay_reason INTO ls_delay_reason.
*      IF sy-tabix = 1.
*        lv_delayed_text = ls_delay_reason-line.
*      ELSE.
*        CONCATENATE lv_delayed_text ls_delay_reason-line INTO lv_delayed_text SEPARATED BY cl_abap_char_utilities=>cr_lf.
*      ENDIF.
*    ENDLOOP.
*  ENDIF.

  IF ls_audit_header-audit_no IS INITIAL.
    CALL FUNCTION 'NUMBER_GET_NEXT'
      EXPORTING
        nr_range_nr             = lv_range_nr
        object                  = lv_object
      IMPORTING
        number                  = lv_audit_number
      EXCEPTIONS
        interval_not_found      = 1
        number_range_not_intern = 2
        object_not_found        = 3
        quantity_is_0           = 4
        quantity_is_not_1       = 5
        interval_overflow       = 6
        buffer_overflow         = 7.
    IF sy-subrc NE 0.
      MESSAGE 'PLEASE MAINTAIN NUMBER RANGE' TYPE 'I'.
```

```
      EXIT.
    ELSE.
*—— ASSIGN NUMBER AT HEADER LEVEL
      ls_audit_header-audit_no = lv_audit_number.
*—— ASSIGN NUMBER AT OBSERVATION LEVEL
      ls_audit_obs-audit_no = lv_audit_number.
*      LOOP AT gt_audit_obs ASSIGNING <fs_obs>.
*        <fs_obs>-audit_no = lv_audit_number.
*      ENDLOOP.
    ENDIF.
  ELSE.
    ls_audit_obs-audit_no = ls_audit_header-audit_no.
*    LOOP AT gt_audit_obs ASSIGNING <fs_obs>.
*      <fs_obs>-audit_no = ls_audit_header-audit_no.
*    ENDLOOP.
  ENDIF.

CLEAR: lt_mod.

  TRY.
      CREATE DATA lr_audit.
*      CLEAR: gs_audit_key.
      CLEAR: ls_text, lt_text.

*      READ TABLE gt_audit_key INTO gs_audit_key WITH KEY audit_no = ls_audit_header-audit_no.
*      IF sy-subrc EQ 0.
*        lr_audit->key            = gs_audit_key-db_key.
*        lr_audit->audit_plant    = ls_audit_data-plant. "gs_audit_key-audit_plant.
*        lr_audit->audit_no       = gs_audit_key-audit_no.
*        lr_audit->audit_type     = ls_audit_data-autype. "gs_audit_key-audit_type.
*        lr_audit->audit_date     = gs_audit_key-audit_date.
*        lr_audit->audit_from     = ls_audit_data-aufrom. "gs_audit_key-audit_from.
*        lr_audit->audit_to       = gs_audit_key-audit_to.
*        lr_audit->audit_rec_date = ls_audit_data-rcdate. "gs_audit_key-audit_rec_date.
*        lr_audit->audit_scope    = gs_audit_key-audit_scope.
*        lr_audit->audit_agency   = gs_audit_key-audit_agency.
*        lr_audit->cname          = gs_audit_key-cname.
*        lr_audit->cdate          = gs_audit_key-cdate.
*        lr_audit->ctime          = gs_audit_key-ctime.
*        lr_audit->delay_reason   = gs_audit_key-delay_reason.
*        lr_audit->leanings       = gs_audit_key-leanings.
*
*        APPEND INITIAL LINE TO lt_mod ASSIGNING <ls_mod>.
*        <ls_mod>-node        = yif_y_ehs_audit1_c=>sc_node-audit_header.
*        <ls_mod>-change_mode = /bobf/if_frw_c=>sc_modify_update.
*        <ls_mod>-key         = lr_audit->key.
*        <ls_mod>-data        = lr_audit.
*      ELSE.
         lr_audit->key = /bobf/cl_frw_factory=>get_new_key( ).
         lr_audit->audit_plant    = ls_audit_data-plant. "gs_audit_header-audit_plant.
         lr_audit->audit_no       = ls_audit_header-audit_no.
         lr_audit->audit_type     = ls_audit_data-autype. "gs_audit_header-audit_type.
         lr_audit->audit_date     = ls_audit_header-audit_date.
```

```
        lr_audit->audit_from     = lv_aufrom. "gs_audit_header-audit_from.
        lr_audit->audit_to       = ls_audit_header-audit_to.
        lr_audit->audit_rec_date = lv_rcdate. "gs_audit_header-audit_rec_date.
        lr_audit->audit_scope    = ls_audit_header-audit_scope.
        lr_audit->audit_agency   = ls_audit_header-audit_agency.
        lr_audit->cname          = sy-uname.
        lr_audit->cdate          = sy-datum.
        lr_audit->ctime          = sy-uzeit.
        lr_audit->delay_reason   = lv_delayed_text.

        APPEND INITIAL LINE TO lt_mod ASSIGNING <ls_mod>.
        <ls_mod>-node        = yif_y_ehs_audit1_c=>sc_node-audit_header.
        <ls_mod>-change_mode = /bobf/if_frw_c=>sc_modify_create.
        <ls_mod>-key         = lr_audit->key.
        <ls_mod>-data        = lr_audit.
*      ENDIF.

*—— SET AUDITORS
      LOOP AT it_auditors ASSIGNING <fs_auditors>.
        FREE lr_auditors.
        CREATE DATA lr_auditors.
        lr_auditors->key = /bobf/cl_frw_factory=>get_new_key( ).
        lr_auditors->auditor = <fs_auditors>-auditor.

        APPEND INITIAL LINE TO lt_mod ASSIGNING <ls_mod>.
        <ls_mod>-node        = yif_y_ehs_audit1_c=>sc_node-auditors.
        <ls_mod>-change_mode = /bobf/if_frw_c=>sc_modify_create.
        <ls_mod>-source_node = yif_y_ehs_audit1_c=>sc_node-audit_header.
        <ls_mod>-association = yif_y_ehs_audit1_c=>sc_association-audit_header-auditors.
        <ls_mod>-source_key  = lr_audit->key.
        <ls_mod>-key         = lr_auditors->key.
        <ls_mod>-data        = lr_auditors.
      ENDLOOP.


*—— SET OBSERVATIONS

  IF ev_coordinator_role = 'X'."gv_observation_text IS NOT INITIAL.
    ls_text-obs_no = ls_audit_obs-obs_no.
    ls_text-type   = 'OBS'.
    ls_text-text   = ls_audit_data-obstext. "gv_observation_text.
      APPEND ls_text TO lt_text.
  ENDIF.

        FREE lr_obs.
        CREATE DATA lr_obs.
        lr_obs->key           = /bobf/cl_frw_factory=>get_new_key( ).
        lr_obs->audit_no      = ls_audit_obs-audit_no.
        lr_obs->obs_no        = ls_audit_obs-obs_no.
        lr_obs->audit_ref_no  = ls_audit_data-refno.
        lr_obs->saved_status  = lv_save_mode.
        lr_obs->location      = ls_audit_data-location.
        lr_obs->sublocation   = ls_audit_data-subloc.
        lr_obs->priority      = ls_audit_data-priority.
        lr_obs->target_date   = lv_tgt_date. "ls_audit_data-tgtdate.
        lr_obs->status        = ls_audit_obs-status.
        lr_obs->resp_person1  = ls_audit_data-impl01.
        lr_obs->resp_person2  = ls_audit_data-impl02.
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        lr_obs->resp_person3  = ls_audit_data-impl03.
*————————————————SOC ADDED BY SHUBHAM DATE:15.06.2020 CHARM:4000002126————————————*
        lr_obs->resp_person4  = ls_audit_data-impl04 .
        lr_obs->resp_person5  = ls_audit_data-impl05 .
        lr_obs->resp_person6  = ls_audit_data-impl06 .
        lr_obs->resp_person7  = ls_audit_data-impl07 .
        lr_obs->resp_person8  = ls_audit_data-impl08 .
        lr_obs->resp_person9  = ls_audit_data-impl09 .
        lr_obs->resp_person10 = ls_audit_data-impl10.
        lr_obs->resp_person11 = ls_audit_data-impl11.
        lr_obs->resp_person12 = ls_audit_data-impl12.
        lr_obs->resp_person13 = ls_audit_data-impl13.
        lr_obs->resp_person14 = ls_audit_data-impl14.
        lr_obs->resp_person15 = ls_audit_data-impl15.
*————————————————EOC ADDED BY SHUBHAM DATE:15.06.2020 CHARM:4000002126————————————*
        lr_obs->cname         = sy-uname.
        lr_obs->cdate         = sy-datum.
        lr_obs->ctime         = sy-uzeit.

        APPEND INITIAL LINE TO lt_mod ASSIGNING <ls_mod>.
        <ls_mod>-node        = yif_y_ehs_audit1_c=>sc_node-audit_observations.
        <ls_mod>-change_mode = /bobf/if_frw_c=>sc_modify_create.
        <ls_mod>-source_node = yif_y_ehs_audit1_c=>sc_node-audit_header.
        <ls_mod>-association = yif_y_ehs_audit1_c=>sc_association-audit_header-audit_observations.
        <ls_mod>-source_key  = lr_audit->key.
        <ls_mod>-key         = lr_obs->key.
        <ls_mod>-data        = lr_obs.

**—— ADD MILESTONES
*        LOOP AT gt_milestone_final INTO gs_milestone_final WHERE obs_no = <fs_obs>-obs_no.
*          FREE lr_milestone.
*          CREATE DATA lr_milestone.
*          lr_milestone->key          = /bobf/cl_frw_factory=>get_new_key( ).
*          lr_milestone->parent_key   = lr_obs->key.
*          lr_milestone->root_key     = lr_audit->key.
*          lr_milestone->audit_no     = <fs_obs>-audit_no.
*          lr_milestone->obs_no       = gs_milestone_final-obs_no.
*          lr_milestone->milestone    = gs_milestone_final-milestone.
*          lr_milestone->ms_target_dt = gs_milestone_final-ms_target_dt.
*          lr_milestone->ms_status    = gs_milestone_final-ms_status.
*          lr_milestone->udate        = sy-datum.
*          lr_milestone->uname        = sy-uname.
**————————————————SOC ADDED BY SHUBHAM DATE:26.06.2020 CHARM:4000002126————————————*
*          lr_milestone->emp_id        = gs_milestone_final-emp_id.
**————————————————SOC ADDED BY SHUBHAM DATE:26.06.2020 CHARM:4000002126————————————*
*
*
*          APPEND INITIAL LINE TO lt_mod ASSIGNING <ls_mod>.
*          <ls_mod>-node        = yif_y_ehs_audit1_c=>sc_node-milestones.
*          <ls_mod>-change_mode = /bobf/if_frw_c=>sc_modify_create.
*          <ls_mod>-source_node = yif_y_ehs_audit1_c=>sc_node-audit_observations.
```

```
*          <ls_mod>-association = yif_y_ehs_audit1_c=>sc_association-audit_observations-milestones.
*          <ls_mod>-source_key  = lr_obs->key.
*          <ls_mod>-root_key    = lr_audit->key.
*          <ls_mod>-key         = lr_milestone->key.
*          <ls_mod>-data        = lr_milestone.
*        ENDLOOP.

*—— SET AUDIT'S OBSERVATION TEXT
        FREE lr_aud_obs.
        CREATE DATA lr_aud_obs.
        READ TABLE lt_text INTO ls_text WITH KEY obs_no = ls_audit_obs-obs_no
                                                 type   = 'OBS'.
        IF sy-subrc EQ 0.
          lr_aud_obs->key        = /bobf/cl_frw_factory=>get_new_key( ).
          lr_aud_obs->created_by = sy-uname.
          lr_aud_obs->created_on = sy-datum.
          lr_aud_obs->audit_text = ls_text-text.

          APPEND INITIAL LINE TO lt_mod ASSIGNING <ls_mod>.
          <ls_mod>-node        = yif_y_ehs_audit1_c=>sc_node-audit_obs_text.
          <ls_mod>-change_mode = /bobf/if_frw_c=>sc_modify_create.
          <ls_mod>-source_node = yif_y_ehs_audit1_c=>sc_node-audit_observations.
          <ls_mod>-association = yif_y_ehs_audit1_c=>sc_association-audit_observations-audit_obs_text.
          <ls_mod>-source_key  = lr_obs->key.
          <ls_mod>-root_key    = lr_audit->key.
          <ls_mod>-key         = lr_aud_obs->key.
          <ls_mod>-data        = lr_aud_obs.
        ENDIF.
*      ENDLOOP.
**SOC BY MANMOHAN/RISHI TR DVRK9A1NH9 DATE: 18/11/2025
**      IF p_gv_save_mode EQ 'S'.
*        SELECT *
*          FROM ydehs_audit_obs
*          INTO TABLE @DATA(it_del_obs)
**          FOR ALL ENTRIES IN @lt_audit_obs1
*          WHERE audit_no EQ @ls_audit_obs-audit_no
*          AND   obs_no   EQ @ls_audit_obs-obs_no
*          AND   saved_status EQ 'D'.
*        IF sy-subrc EQ 0.
*          LOOP AT it_del_obs INTO DATA(wa_del_obs).
*            DELETE FROM ydehs_audit_obs WHERE audit_no EQ wa_del_obs-audit_no AND obs_no EQ wa_del_obs-obs_no AND saved_status EQ 'D'.
*          ENDLOOP.
*        ENDIF.
**      ENDIF.
**EOC BY MANMOHAN/RISHI TR DVRK9A1NH9 DATE: 18/11/2025
*—— CREATE RECORD
      CALL METHOD mo_svc_mngr->modify
        EXPORTING
          it_modification = lt_mod
        IMPORTING
          eo_change       = lo_change
          eo_message      = lo_message.

lv_row += 1.
*——— CHECK FOR ERRORS:
```

```
      IF lo_message IS BOUND.
        IF lo_message->check( ) EQ abap_true.
          lo_message->get_messages( IMPORTING et_message = lt_messages ).
          LOOP AT lt_messages ASSIGNING <ls_message>.
            IF sy-tabix = 1.
            ls_return-row = lv_row.
            ls_return-message = <ls_message>-message->get_text( ).
            APPEND ls_return to lt_return.
            ENDIF.
*            gv_msg_text = <ls_message>-message->get_text( ).
*            WRITE: / gv_msg_text.
          ENDLOOP.
          CLEAR: lt_messages.
          EXIT.
        ENDIF.
      ENDIF.

*—— APPLY THE TRANSACTIONAL CHANGES:
      CALL METHOD mo_txn_mngr->save
        IMPORTING
          eo_message  = lo_message
          ev_rejected = lv_rejected.

*—— CHECK MESSAGES
      IF lv_rejected EQ abap_true.
        IF lo_message->check( ) EQ abap_true.
          lo_message->get_messages( IMPORTING et_message = lt_messages ).
          LOOP AT lt_messages ASSIGNING <ls_message>.
            IF sy-tabix = 1.
            ls_return-row = lv_row.
            ls_return-message = <ls_message>-message->get_text( ).
            APPEND ls_return to lt_return.
            ENDIF.
*            gv_msg_text = <ls_message>-message->get_text( ).
*            WRITE: / gv_msg_text.
          ENDLOOP.
          CLEAR: lt_messages.
          EXIT.
        ENDIF.
      ELSE.

*          LOOP AT gt_audit_obs INTO DATA(fs_audit_obs).
            ls_audit_chang-audit_no = ls_audit_header-audit_no.
            ls_audit_chang-obs_no = ls_audit_obs-obs_no.
            ls_audit_chang-status = ls_audit_obs-status.
            ls_audit_chang-change_on = sy-datum.
            ls_audit_chang-change_time = sy-uzeit.
            ls_audit_chang-change_by = sy-uname.
            APPEND ls_audit_chang TO lt_audit_chang.
*            MODIFY yehs_audit_chang FROM ls_audit_chang.
*          ENDLOOP.
        ENDIF.

    CATCH /bobf/cx_frw INTO lx_bopf_ex.
      lv_err_msg = lx_bopf_ex->get_text( ).
*      WRITE: / lv_err_msg.
      MESSAGE : lv_err_msg TYPE 'E'.
  ENDTRY.

```

```
CLEAR: ls_audit_obs-audit_no, ls_audit_data, ls_return, ls_audit_chang,
ls_audit_header-audit_no.

ENDLOOP.

*        INSERT INTO YEHS_AUDIT_CHANG VALUES ls_audit_chang.
            MODIFY yehs_audit_chang FROM TABLE lt_audit_chang.

ENDFUNCTION.

FORM date_conv USING p_date TYPE char20 CHANGING lv_date.

DATA: lv_day   TYPE string,
      lv_mon   TYPE string,
      lv_year  TYPE string.
*      lv_date  TYPE sy-datum.

SPLIT p_date AT '-' INTO lv_day lv_mon lv_year.
CASE lv_mon.
  WHEN 'Jan'. lv_mon = '01'.
  WHEN 'Feb'. lv_mon = '02'.
  WHEN 'Mar'. lv_mon = '03'.
  WHEN 'Apr'. lv_mon = '04'.
  WHEN 'May'. lv_mon = '05'.
  WHEN 'Jun'. lv_mon = '06'.
  WHEN 'Jul'. lv_mon = '07'.
  WHEN 'Aug'. lv_mon = '08'.
  WHEN 'Sep'. lv_mon = '09'.
  WHEN 'Oct'. lv_mon = '10'.
  WHEN 'Nov'. lv_mon = '11'.
  WHEN 'Dec'. lv_mon = '12'.
ENDCASE.
IF lv_day < 10.
  CONCATENATE '0' lv_day INTO lv_day.
ENDIF.
CONCATENATE sy-datum+0(2) lv_year lv_mon lv_day INTO lv_date.
CLEAR: lv_year, lv_mon, lv_day.
  ENDFORM.
```

Subroutines:

		Line	Page
DATE_CONV.....	LYEHSM_EXCEL_DOWNLOADU04	160	3
	LYEHSM_EXCEL_DOWNLOADU04	170	3
	LYEHSM_EXCEL_DOWNLOADU04	518	10

Messages:

		Line	Page
?.....	LYEHSM_EXCEL_DOWNLOADU04	232	4
?.....	LYEHSM_EXCEL_DOWNLOADU04	506	9
YE01030.....	LYEHSM_EXCEL_DOWNLOADU04	106	2
	LYEHSM_EXCEL_DOWNLOADU04	116	2
	LYEHSM_EXCEL_DOWNLOADU04	130	3
	LYEHSM_EXCEL_DOWNLOADU04	141	3

External calls

Function Modules		Line	Page
CONVERT_DATE_INPUT.....	LYEHSM_EXCEL_DOWNLOADU04	181	4
GET_DOMAIN_VALUES.....	LYEHSM_EXCEL_DOWNLOADU04	92	2
NUMBER_GET_NEXT.....	LYEHSM_EXCEL_DOWNLOADU04	217	4

DATABASE OPERATIONS

MODIFY:		Line	Page
YEHS_AUDIT_CHANG.....	LYEHSM_EXCEL_DOWNLOADU04	514	10

INTERNAL TABLE OPERATIONS

READ TABLE:		Line	Page
LT_STATUS.....	LYEHSM_EXCEL_DOWNLOADU04	99	2
LT_TEXT.....	LYEHSM_EXCEL_DOWNLOADU04	405	8

```
FUNCTION YEHSM_ECEL_DATA_VALID.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     REFERENCE(IS_AUDIT_HEADER) TYPE  YSROOT1_D
*"  EXPORTING
*"     REFERENCE(EV_SAVE) TYPE  FLAG
*"  TABLES
*"      IT_EXCEL_DATA TYPE  YTTEHS_EXCEL_D
*"----------------------------------------------------------------------
*TYPES: BEGIN OF ty_upload,
*         sno       TYPE i,             " S No.
*         plant     TYPE werks_d,       " Plant
*         autype    TYPE char40,        " Type of Audit/Activity
*         aufrom    TYPE char20,        " Audit/Activity From
*         rcdate    TYPE char20,        " Audit Report received date
*         refno     TYPE char40,        " Reference No
*         priority  TYPE char10,        " Priority (L/M/H)
*         tgtdate   TYPE char20,        " Target Date
*         location  TYPE char40,        " Location
*         subloc    TYPE char40,        " Sub Location
*         obstext   TYPE char1024,      " Observation Text
*         impl01    TYPE char10,        " Implementor ID 1
*         impl02    TYPE char10,        " Implementor ID 2
*         impl03    TYPE char10,        " Implementor ID 3
*         impl04    TYPE char10,        " Implementor ID 4
*         impl05    TYPE char10,        " Implementor ID 5
*         impl06    TYPE char10,        " Implementor ID 6
*         impl07    TYPE char10,        " Implementor ID 7
*         impl08    TYPE char10,        " Implementor ID 8
*         impl09    TYPE char10,        " Implementor ID 9
*         impl10    TYPE char10,        " Implementor ID 10
*         impl11    TYPE char10,        " Implementor ID 11
*         impl12    TYPE char10,        " Implementor ID 12
*         impl13    TYPE char10,        " Implementor ID 13
*         impl14    TYPE char10,        " Implementor ID 14
*         impl15    TYPE char10,        " Implementor ID 15
*       END OF ty_upload.



DATA: lv_rcdate TYPE datum,
      lv_aufrom TYPE datum,
      ls_audit_header TYPE YSROOT1_D.
*      lt_audit_data   TYPE YTTEHS_EXCEL_D.
*      lv_flag TYPE flag.
ls_audit_header = is_audit_header.

LOOP AT it_excel_data INTO DATA(ls_upload).

CLEAR: lv_rcdate, lv_aufrom.
  IF ls_upload-impl01 IS INITIAL OR ls_upload-obstext IS INITIAL
     OR ls_upload-refno IS INITIAL OR ls_upload-tgtdate IS INITIAL OR ls_
upload-priority IS INITIAL
     OR ls_upload-rcdate IS INITIAL OR ls_upload-autype IS INITIAL.
       ev_save = abap_true.
       MESSAGE : 'Excel is incomplete : enter all the fields' TYPE 'E'.
       EXIT.
  ELSE.
  CALL FUNCTION 'CONVERT_DATE_INPUT'
  EXPORTING
```

```
    input  = ls_upload-rcdate
  IMPORTING
    output = lv_rcdate
  EXCEPTIONS
    OTHERS = 1.
  IF sy-subrc <> 0.
      MESSAGE 'Invalid RC Date format' TYPE 'E'.
      EXIT.
    ENDIF.
      IF lv_rcdate GT sy-datum.
        ev_save = abap_true.
    MESSAGE 'AUDIT/ACTIVITY REPORT RECEIVED DATE CANNOT BE IN FUTURE' TYPE 'E'.
    EXIT.
  ENDIF.

  CALL FUNCTION 'CONVERT_DATE_INPUT'
  EXPORTING
    input  = ls_upload-rcdate
  IMPORTING
    output = lv_aufrom
  EXCEPTIONS
    OTHERS = 1.
    IF lv_aufrom GT sy-datum.
      ev_save = abap_true.
    MESSAGE 'AUDIT/ACTIVITY FROM DATE CAN NOT BE IN FUTURE' TYPE 'E'.
    EXIT.
  ENDIF.

    IF lv_aufrom GT ls_audit_header-audit_to. "gs_audit_header-audit_from GT gs_audit_header-audit_to.
      ev_save = abap_true.
    MESSAGE 'AUDIT/ACTIVITY FROM-DATE CAN NOT BE GREATER THAN AUDIT-TO DATE' TYPE 'E'.
    EXIT.
  ENDIF.
    ENDIF.

    CLEAR ls_upload.
ENDLOOP.

*gs_audit_obs-obs_no = 1.
*gs_audit_obs-status = 'O'.

ENDFUNCTION.
```

Messages:

		Line	Page
?.....	LYEHSM_EXCEL_DOWNLOADU03	56	15
?.....	LYEHSM_EXCEL_DOWNLOADU03	67	16
?.....	LYEHSM_EXCEL_DOWNLOADU03	72	16
?.....	LYEHSM_EXCEL_DOWNLOADU03	85	16
?.....	LYEHSM_EXCEL_DOWNLOADU03	91	16

External calls

Function Modules		Line	Page
CONVERT_DATE_INPUT.....	LYEHSM_EXCEL_DOWNLOADU03	59	15
	LYEHSM_EXCEL_DOWNLOADU03	76	16

```
FUNCTION YEHSM_ECEL_DOWN.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(WERKS) TYPE  WERKS_D OPTIONAL
*"     VALUE(AUTYP) TYPE  CHAR40 OPTIONAL
*"     VALUE(STDAT) TYPE  DATUM OPTIONAL
*"     VALUE(RCDAT) TYPE  DATUM OPTIONAL
*"     VALUE(OBSNO) TYPE  I DEFAULT 10
*"----------------------------------------------------------------------


p_werks = WERKS.
p_autyp = AUTYP.
p_stdat = STDAT.
p_rcdat = RCDAT.
p_obsno = OBSNO.


* Fetch locations from YEHS_LOCATIONS for the selected plant
  SELECT plant location sub_location
        FROM yehs_locations
    INTO TABLE lt_locations
    WHERE plant = p_werks.

* Build the location dropdown string (comma-separated)
  DATA: lv_loc_dropdown TYPE string.
  LOOP AT lt_locations INTO DATA(ls_loc).
    IF lv_loc_dropdown IS INITIAL.
      lv_loc_dropdown = ls_loc-locid && | | && ls_loc-locdesc.
    ELSE.
      lv_loc_dropdown = lv_loc_dropdown && |,| && ls_loc-locid && | | &&
 ls_loc-locdesc.
    ENDIF.
  ENDLOOP.

  IF lv_loc_dropdown IS INITIAL.
    lv_loc_dropdown = ' '.
  ENDIF.

* Format dates for Excel (d-mmm-yyyy)
  DATA: lv_stdate TYPE string,
        lv_rcdate TYPE string.

  PERFORM format_date USING p_stdat CHANGING lv_stdate.
  PERFORM format_date USING p_rcdat CHANGING lv_rcdate.

* Build the Excel XML
  PERFORM build_excel_xml USING p_werks
                                p_autyp
                                lv_stdate
                                lv_rcdate
                                p_obsno
                                lv_loc_dropdown
                        CHANGING lv_xml.

* Convert string to xstring
  CALL FUNCTION 'SCMS_STRING_TO_XSTRING'
    EXPORTING
      text   = lv_xml
    IMPORTING
```

```
      buffer = lv_xstring
    EXCEPTIONS
      OTHERS = 1.

  IF sy-subrc <> 0.
    MESSAGE 'Error converting XML string' TYPE 'E'.
    RETURN.
  ENDIF.

* File save dialog - .xls extension for direct Excel open
  CALL METHOD cl_gui_frontend_services=>file_save_dialog
    EXPORTING
      default_extension = 'xls'
      default_file_name = |EHSM_Audit_{ p_werks }_{ sy-datum }.xls|
      file_filter       = 'Excel Files (*.xls)|*.xls'
    CHANGING
      filename          = lv_filename
      path              = lv_path
      fullpath          = lv_fullpath
      user_action       = lv_action.

  IF lv_action <> cl_gui_frontend_services=>action_ok.
    MESSAGE 'Download cancelled by user' TYPE 'S'.
    RETURN.
  ENDIF.

* Convert to binary
  CALL FUNCTION 'SCMS_XSTRING_TO_BINARY'
    EXPORTING
      buffer        = lv_xstring
    IMPORTING
      output_length = lv_size
    TABLES
      binary_tab    = lt_binary.

* Download
  CALL METHOD cl_gui_frontend_services=>gui_download
    EXPORTING
      bin_filesize = lv_size
      filename     = lv_fullpath
      filetype     = 'BIN'
    CHANGING
      data_tab     = lt_binary
    EXCEPTIONS
      OTHERS       = 1.

*  IF sy-subrc = 0.
*    WRITE: / |File downloaded successfully: { lv_fullpath }|.
*    WRITE: / |Plant: { p_werks }|.
*    WRITE: / |Audit Type: { p_autyp }|.
*    WRITE: / |Start Date: { lv_stdate }|.
*    WRITE: / |Received Date: { lv_rcdate }|.
*    WRITE: / |Total Observations (Rows): { p_obsno }|.
*    MESSAGE |File downloaded successfully to { lv_fullpath }| TYPE 'S'.
*  ELSE.
*    MESSAGE 'Error downloading file' TYPE 'E'.
*  ENDIF.



ENDFUNCTION.
```

Subroutines:

		Line	Page
BUILD_EXCEL_XML.....	LYEHSM_EXCEL_DOWNLOADU01	48	19
FORMAT_DATE.....	LYEHSM_EXCEL_DOWNLOADU01	44	19
	LYEHSM_EXCEL_DOWNLOADU01	45	19

Messages:

		Line	Page
?.....	LYEHSM_EXCEL_DOWNLOADU01	66	20
?.....	LYEHSM_EXCEL_DOWNLOADU01	83	20

External calls

Function Modules		Line	Page
SCMS_STRING_TO_XSTRING.....	LYEHSM_EXCEL_DOWNLOADU01	57	19
SCMS_XSTRING_TO_BINARY.....	LYEHSM_EXCEL_DOWNLOADU01	88	20

DATABASE OPERATIONS

SELECT:		Line	Page
YEHS_LOCATIONS.....	LYEHSM_EXCEL_DOWNLOADU01	22	19

```
FUNCTION YEHSM_ECEL_UPLOAD.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  TABLES
*"      IT_EXCEL_DATA TYPE  YTTEHS_EXCEL_D OPTIONAL
*"----------------------------------------------------------------------
data p_file TYPE rlgrap-filename.

*----------------------------------------------------------------------*
* Types
*----------------------------------------------------------------------*
TYPES: BEGIN OF ty_upload,
         sno       TYPE i,             " S No.
         plant     TYPE werks_d,       " Plant
         autype    TYPE char40,        " Type of Audit/Activity
         aufrom    TYPE char20,        " Audit/Activity From
         rcdate    TYPE char20,        " Audit Report received date
         refno     TYPE char40,        " Reference No
         priority  TYPE char10,        " Priority (L/M/H)
         tgtdate   TYPE char20,        " Target Date
         location  TYPE char40,        " Location
         subloc    TYPE char40,        " Sub Location
         obstext   TYPE char1024,      " Observation Text
         impl01    TYPE char10,        " Implementor ID 1
         impl02    TYPE char10,        " Implementor ID 2
         impl03    TYPE char10,        " Implementor ID 3
         impl04    TYPE char10,        " Implementor ID 4
         impl05    TYPE char10,        " Implementor ID 5
         impl06    TYPE char10,        " Implementor ID 6
         impl07    TYPE char10,        " Implementor ID 7
         impl08    TYPE char10,        " Implementor ID 8
         impl09    TYPE char10,        " Implementor ID 9
         impl10    TYPE char10,        " Implementor ID 10
         impl11    TYPE char10,        " Implementor ID 11
         impl12    TYPE char10,        " Implementor ID 12
         impl13    TYPE char10,        " Implementor ID 13
         impl14    TYPE char10,        " Implementor ID 14
         impl15    TYPE char10,        " Implementor ID 15
       END OF ty_upload.

TYPES: BEGIN OF ty_raw,
         row TYPE i,
         col TYPE i,
         val TYPE char1024,
       END OF ty_raw.

DATA: lt_upload   TYPE TABLE OF ty_upload,
      ls_upload   TYPE ty_upload,
      lt_raw      TYPE TABLE OF ty_raw,
      ls_raw      TYPE ty_raw,
      lt_intern   TYPE TABLE OF alsmex_tabline,
      ls_intern   TYPE alsmex_tabline,
      lv_file     TYPE rlgrap-filename,
      lv_rc       TYPE i.

  DATA: lt_filetab TYPE filetable,
        lv_subrc   TYPE i.

  CALL METHOD cl_gui_frontend_services=>file_open_dialog
    EXPORTING
      window_title = 'Select Excel File'
```

```
      file_filter  = 'Excel Files (*.xls;*.xlsx)|*.xls;*.xlsx'
    CHANGING
      file_table   = lt_filetab
      rc           = lv_subrc.

  IF lv_subrc = 1.
    READ TABLE lt_filetab INTO DATA(ls_file) INDEX 1.
    IF sy-subrc = 0.
      p_file = ls_file-filename.
    ENDIF.
  ENDIF.

* Upload Excel using ALSM_EXCEL_TO_INTERNAL_TABLE
  CALL FUNCTION 'ALSM_EXCEL_TO_INTERNAL_TABLE'
    EXPORTING
      filename                = p_file
      i_begin_col             = 1
      i_begin_row             = 2        " Skip header row (Row 1)
      i_end_col               = 26       " Columns A to Z
      i_end_row               = 9999
    TABLES
      intern                  = lt_intern
    EXCEPTIONS
      inconsistent_parameters = 1
      upload_ole              = 2
      OTHERS                  = 3.

  IF sy-subrc <> 0.
    MESSAGE |Error uploading Excel file. SY-SUBRC = { sy-subrc }| TYPE 'E'.
    RETURN.
  ENDIF.

  IF lt_intern IS INITIAL.
    MESSAGE 'No data found in Excel file' TYPE 'E'.
    RETURN.
  ENDIF.

* Convert ALSM format to raw table
  LOOP AT lt_intern INTO ls_intern.
    ls_raw-row = ls_intern-row.
    ls_raw-col = ls_intern-col.
    ls_raw-val = ls_intern-value.
    APPEND ls_raw TO lt_raw.
  ENDLOOP.

* Build internal table from raw data
  DATA: lv_prev_row TYPE i VALUE 0.

  SORT lt_raw BY row col.

  LOOP AT lt_raw INTO ls_raw.

*   New row - save previous and clear
    IF lv_prev_row <> 0 AND ls_raw-row <> lv_prev_row.
      IF ls_upload-plant IS NOT INITIAL.  " Skip empty rows
        APPEND ls_upload TO lt_upload.
      ENDIF.
      CLEAR ls_upload.
    ENDIF.

```

```
      lv_prev_row = ls_raw-row.

*   Map columns to structure fields
      CASE ls_raw-col.
        WHEN 1.  ls_upload-sno      = ls_raw-val.
        WHEN 2.  ls_upload-plant    = ls_raw-val.
        WHEN 3.  ls_upload-autype   = ls_raw-val.
        WHEN 4.  ls_upload-aufrom   = ls_raw-val.
        WHEN 5.  ls_upload-rcdate   = ls_raw-val.
        WHEN 6.  ls_upload-refno    = ls_raw-val.
        WHEN 7.  ls_upload-priority = ls_raw-val.
        WHEN 8.  ls_upload-tgtdate  = ls_raw-val.
        WHEN 9.  ls_upload-location = ls_raw-val.
        WHEN 10. ls_upload-subloc   = ls_raw-val.
        WHEN 11. ls_upload-obstext  = ls_raw-val.
        WHEN 12. ls_upload-impl01   = ls_raw-val.
        WHEN 13. ls_upload-impl02   = ls_raw-val.
        WHEN 14. ls_upload-impl03   = ls_raw-val.
        WHEN 15. ls_upload-impl04   = ls_raw-val.
        WHEN 16. ls_upload-impl05   = ls_raw-val.
        WHEN 17. ls_upload-impl06   = ls_raw-val.
        WHEN 18. ls_upload-impl07   = ls_raw-val.
        WHEN 19. ls_upload-impl08   = ls_raw-val.
        WHEN 20. ls_upload-impl09   = ls_raw-val.
        WHEN 21. ls_upload-impl10   = ls_raw-val.
        WHEN 22. ls_upload-impl11   = ls_raw-val.
        WHEN 23. ls_upload-impl12   = ls_raw-val.
        WHEN 24. ls_upload-impl13   = ls_raw-val.
        WHEN 25. ls_upload-impl14   = ls_raw-val.
        WHEN 26. ls_upload-impl15   = ls_raw-val.
      ENDCASE.

    ENDLOOP.

* Append last row
    IF ls_upload-plant IS NOT INITIAL.
      APPEND ls_upload TO lt_upload.
    ENDIF.

* Append last row
    IF ls_upload-plant IS NOT INITIAL.
      APPEND ls_upload TO gt_upload.
    ENDIF.

* Display count
    DATA(lv_count) = lines( gt_upload ).

* Display data in ALV
    IF lt_upload IS NOT INITIAL.
      data ls_YTTEHS_EXCEL_D type YSEHS_EXCEL_D.
*     PERFORM display_alv USING lv_count.
      loop at lt_upload INTO ls_upload.
        MOVE-CORRESPONDING ls_upload to ls_YTTEHS_EXCEL_D.
        append ls_YTTEHS_EXCEL_D to it_excel_data.
        ENDLOOP.
    ELSE.
      MESSAGE 'No valid data found in Excel file' TYPE 'S' DISPLAY LIKE 'E'.
    ENDIF.


```

```

ENDFUNCTION.
```

Messages:

		Line	Page
.....	LYEHSM_EXCEL_DOWNLOADU02	90	25
?.....	LYEHSM_EXCEL_DOWNLOADU02	95	25
?.....	LYEHSM_EXCEL_DOWNLOADU02	178	26

External calls

Function Modules		Line	Page
ALSM_EXCEL_TO_INTERNAL_TABLE..	LYEHSM_EXCEL_DOWNLOADU02	75	25

INTERNAL TABLE OPERATIONS

READ TABLE:		Line	Page
LT_FILETAB.....	LYEHSM_EXCEL_DOWNLOADU02	68	25

```
FUNCTION-POOL YEHSM_EXCEL_DOWNLOAD.         "MESSAGE-ID ..

* INCLUDE LYEHSM_EXCEL_DOWNLOADD...          " Local class definition

*____________________________________________________________________*
* Types & Data Declarations
*____________________________________________________________________*
TYPES: BEGIN OF ty_location,
         werks   TYPE werks_d,
         locid   TYPE char40,
         locdesc TYPE char80,
       END OF ty_location.

TYPES: BEGIN OF ty_audit_type,
         autyp TYPE char40,
         descr TYPE char80,
       END OF ty_audit_type.

DATA: lt_locations  TYPE TABLE OF ty_location,
      lt_audit_type TYPE TABLE OF ty_audit_type,
      lv_xml        TYPE string,
      lv_xstring    TYPE xstring,
      lv_filename   TYPE string,
      lv_fullpath   TYPE string,
      lv_path       TYPE string,
      lv_action     TYPE i,
      lt_binary     TYPE solix_tab,
      lv_size       TYPE i.


data : p_werks  TYPE werks_d   ,  " Plant
              p_autyp  TYPE char40    ,   " Type of Audit/Activity
              p_stdat  TYPE sy-datum  ,   " Audit/Activity Start Date
              p_rcdat  TYPE sy-datum  ,   " Audit Report Received Date
              p_obsno  TYPE i         value 10.   " Total No. of Observa
tions

TYPES: BEGIN OF ty_upload,
         sno       TYPE i,            " S No.
         plant     TYPE werks_d,      " Plant
         autype    TYPE char40,       " Type of Audit/Activity
         aufrom    TYPE char20,       " Audit/Activity From
         rcdate    TYPE char20,       " Audit Report received date
         refno     TYPE char40,       " Reference No
         priority  TYPE char10,       " Priority (L/M/H)
         tgtdate   TYPE char20,       " Target Date
         location  TYPE char40,       " Location
         subloc    TYPE char40,       " Sub Location
         obstext   TYPE char1024,     " Observation Text
         impl01    TYPE char10,       " Implementor ID 1
         impl02    TYPE char10,       " Implementor ID 2
         impl03    TYPE char10,       " Implementor ID 3
         impl04    TYPE char10,       " Implementor ID 4
         impl05    TYPE char10,       " Implementor ID 5
         impl06    TYPE char10,       " Implementor ID 6
         impl07    TYPE char10,       " Implementor ID 7
         impl08    TYPE char10,       " Implementor ID 8
         impl09    TYPE char10,       " Implementor ID 9
         impl10    TYPE char10,       " Implementor ID 10
         impl11    TYPE char10,       " Implementor ID 11
         impl12    TYPE char10,       " Implementor ID 12
```

```
          impl13    TYPE char10,       " Implementor ID 13
          impl14    TYPE char10,       " Implementor ID 14
          impl15    TYPE char10,       " Implementor ID 15
        END OF ty_upload.

TYPES: BEGIN OF ty_raw,
          row TYPE i,
          col TYPE i,
          val TYPE char1024,
        END OF ty_raw.

DATA: gt_upload   TYPE TABLE OF ty_upload,
      ls_upload   TYPE ty_upload,
      lt_raw      TYPE TABLE OF ty_raw,
      ls_raw      TYPE ty_raw,
      lt_intern   TYPE TABLE OF alsmex_tabline,
      ls_intern   TYPE alsmex_tabline,
      lv_file     TYPE rlgrap-filename,
      lv_rc       TYPE i,
      go_alv      TYPE REF TO cl_salv_table.


*————————————————————————————————————————————————————————————————————*
* Class for ALV Event Handling
*————————————————————————————————————————————————————————————————————*
CLASS lcl_event_handler DEFINITION.
  PUBLIC SECTION.
    METHODS: on_user_command FOR EVENT added_function OF cl_salv_events
                 IMPORTING e_salv_function.
ENDCLASS.

CLASS lcl_event_handler IMPLEMENTATION.
  METHOD on_user_command.
    CASE e_salv_function.
      WHEN 'SAVE_DRAFT'.
        PERFORM save_as_draft.
      WHEN 'SAVE_FINAL'.
        PERFORM save_final.
    ENDCASE.
  ENDMETHOD.
ENDCLASS.
*&————————————————————————————————————————————————————————————————————*
*& Form FORMAT_DATE
*&————————————————————————————————————————————————————————————————————*
FORM format_date USING iv_date    TYPE sy-datum
                 CHANGING cv_date TYPE string.

  DATA: lv_day   TYPE string,
        lv_month TYPE string,
        lv_year  TYPE string.

  lv_day  = iv_date+6(2).
  lv_year = iv_date+0(4).

  CASE iv_date+4(2).
    WHEN '01'. lv_month = 'Jan'.
    WHEN '02'. lv_month = 'Feb'.
    WHEN '03'. lv_month = 'Mar'.
    WHEN '04'. lv_month = 'Apr'.
    WHEN '05'. lv_month = 'May'.
    WHEN '06'. lv_month = 'Jun'.
```

```
    WHEN '07'. lv_month = 'Jul'.
    WHEN '08'. lv_month = 'Aug'.
    WHEN '09'. lv_month = 'Sep'.
    WHEN '10'. lv_month = 'Oct'.
    WHEN '11'. lv_month = 'Nov'.
    WHEN '12'. lv_month = 'Dec'.
  ENDCASE.

  cv_date = |{ lv_day }-{ lv_month }-{ lv_year }|.

ENDFORM.

*&---------------------------------------------------------------------*
*& Form BUILD_EXCEL_XML
*&---------------------------------------------------------------------*
FORM build_excel_xml USING iv_plant    TYPE werks_d
                           iv_autype   TYPE char40
                           iv_stdate   TYPE string
                           iv_rcdate   TYPE string
                           iv_rows     TYPE i
                           iv_loc_dd   TYPE string
                   CHANGING cv_xml     TYPE string.

  DATA: lv_row_range   TYPE string,
        lv_loc_range   TYPE string,
        lv_data_rows   TYPE string,
        lv_row_num     TYPE i,
        lv_exp_rows    TYPE i,
        lv_quot        TYPE c VALUE '"'.  " Double-quote character for XML

* Row range: Header=Row1, Data starts Row2 to Row1+iv_rows
  lv_row_num  = 1 + iv_rows.
  lv_exp_rows = 1 + iv_rows.
  lv_row_range = |R2C7:R{ lv_row_num }C7|.
  lv_loc_range = |R2C9:R{ lv_row_num }C9|.

* Build priority dropdown value with proper XML encoding
* Result: "L Low,M Medium,H High"  (with actual quote chars)
  DATA: lv_priority_val TYPE string,
        lv_location_val TYPE string.

  CONCATENATE lv_quot 'L Low,M Medium,H High' lv_quot INTO lv_priority_val.
  CONCATENATE lv_quot iv_loc_dd lv_quot INTO lv_location_val.

* Build data rows
  DATA(lv_sno) = 0.
  DO iv_rows TIMES.
    lv_sno = lv_sno + 1.
    lv_data_rows = lv_data_rows &&
      |   <Row ss:AutoFitHeight="0" ss:Height="120">| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell ss:StyleID="s71"><Data ss:Type="Number">{ lv_sno }</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">{ iv_plant }</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">{ iv_autype }</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">{ iv_stdate }</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
```

```
176         |     <Cell><Data ss:Type="String">{ iv_rcdate }</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
177         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
178         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
179         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
180         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
181         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
182         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
183         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
184         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
185         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
186         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
187         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
188         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
189         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
190         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
191         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
192         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
193         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
194         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
195         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
196         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
197         |     <Cell/>| && cl_abap_char_utilities=>cr_lf &&
198         |   </Row>| && cl_abap_char_utilities=>cr_lf.
199     ENDDO.
200
201   * ==================================================================
202   * Assemble full XML
203   * ==================================================================
204     cv_xml =
205       |<?xml version="1.0"?>| && cl_abap_char_utilities=>cr_lf &&
206       |<?mso-application progid="Excel.Sheet"?>| && cl_abap_char_utilities=>cr_lf &&
207       |<Workbook xmlns="urn:schemas-microsoft-com:office:spreadsheet"| && cl_abap_char_utilities=>cr_lf &&
208       | xmlns:o="urn:schemas-microsoft-com:office:office"| && cl_abap_char_utilities=>cr_lf &&
209       | xmlns:x="urn:schemas-microsoft-com:office:excel"| && cl_abap_char_utilities=>cr_lf &&
210       | xmlns:ss="urn:schemas-microsoft-com:office:spreadsheet"| && cl_abap_char_utilities=>cr_lf &&
211       | xmlns:html="http://www.w3.org/TR/REC-html40">| && cl_abap_char_utilities=>cr_lf &&
212
213       | <DocumentProperties xmlns="urn:schemas-microsoft-com:office:office">| && cl_abap_char_utilities=>cr_lf &&
214       |  <Author>{ sy-uname }</Author>| && cl_abap_char_utilities=>cr_lf &&
215       |  <Created>{ sy-datum+0(4) }-{ sy-datum+4(2) }-{ sy-datum+6(2) }T{ sy-uzeit+0(2) }:{ sy-uzeit+2(2) }:{ sy-uzeit+4(2) }Z</Created>| && cl_abap_char_utilities=>cr_lf &&
216       |  <Version>16.00</Version>| && cl_abap_char_utilities=>cr_lf &&
217       | </DocumentProperties>| && cl_abap_char_utilities=>cr_lf &&
218
219       | <ExcelWorkbook xmlns="urn:schemas-microsoft-com:office:excel">| && cl_abap_char_utilities=>cr_lf &&
220       |  <WindowHeight>7830</WindowHeight>| && cl_abap_char_utilities=>cr_lf &&
221       |  <WindowWidth>19200</WindowWidth>| && cl_abap_char_utilities=>cr_lf &&
222       |  <ProtectStructure>False</ProtectStructure>| && cl_abap_char_utili
```

```
ties=>cr_lf &&
    |  <ProtectWindows>False</ProtectWindows>| && cl_abap_char_utilities=>cr_lf &&
    | </ExcelWorkbook>| && cl_abap_char_utilities=>cr_lf &&

* ===================== STYLES =====================
    | <Styles>| && cl_abap_char_utilities=>cr_lf &&

    |  <Style ss:ID="Default" ss:Name="Normal">| && cl_abap_char_utilities=>cr_lf &&
    |   <Alignment ss:Vertical="Bottom"/>| && cl_abap_char_utilities=>cr_lf &&
    |   <Font ss:FontName="Aptos Narrow" x:Family="Swiss" ss:Size="11" ss:Color="#000000"/>| && cl_abap_char_utilities=>cr_lf &&
    |  </Style>| && cl_abap_char_utilities=>cr_lf &&

    |  <Style ss:ID="s62">| && cl_abap_char_utilities=>cr_lf &&
    |   <Borders>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Bottom" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Left" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Right" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Top" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |   </Borders>| && cl_abap_char_utilities=>cr_lf &&
    |  </Style>| && cl_abap_char_utilities=>cr_lf &&

    |  <Style ss:ID="s63">| && cl_abap_char_utilities=>cr_lf &&
    |   <Alignment ss:Vertical="Top" ss:WrapText="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |   <Borders>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Bottom" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Left" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Right" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Top" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |   </Borders>| && cl_abap_char_utilities=>cr_lf &&
    |   <Font ss:FontName="Aptos Narrow" x:Family="Swiss" ss:Size="11" ss:Color="#000000"/>| && cl_abap_char_utilities=>cr_lf &&
    |   <NumberFormat ss:Format="@"/>| && cl_abap_char_utilities=>cr_lf &&
    |  </Style>| && cl_abap_char_utilities=>cr_lf &&

    |  <Style ss:ID="s64">| && cl_abap_char_utilities=>cr_lf &&
    |   <Alignment ss:Vertical="Top" ss:WrapText="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |   <Borders>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Bottom" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Left" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Right" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |    <Border ss:Position="Top" ss:LineStyle="Continuous" ss:Weight="1"/>| && cl_abap_char_utilities=>cr_lf &&
    |   </Borders>| && cl_abap_char_utilities=>cr_lf &&
```

```
      |    <Font ss:FontName="Aptos Narrow" x:Family="Swiss" ss:Size="11" s
s:Color="#000000"/>| && cl_abap_char_utilities=>cr_lf &&
      |    <NumberFormat ss:Format="[ENG][$-409]d\\-mmm\\-yyyy;@"/>| && cl_
abap_char_utilities=>cr_lf &&
      |  </Style>| && cl_abap_char_utilities=>cr_lf &&

      |  <Style ss:ID="s70">| && cl_abap_char_utilities=>cr_lf &&
      |    <Alignment ss:Vertical="Bottom" ss:WrapText="1"/>| && cl_abap_ch
ar_utilities=>cr_lf &&
      |    <Borders>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Bottom" ss:LineStyle="Continuous" ss:Weigh
t="1"/>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Left" ss:LineStyle="Continuous" ss:Weight=
"1"/>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Right" ss:LineStyle="Continuous" ss:Weight
="1"/>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Top" ss:LineStyle="Continuous" ss:Weight="
1"/>| && cl_abap_char_utilities=>cr_lf &&
      |    </Borders>| && cl_abap_char_utilities=>cr_lf &&
      |    <Font ss:FontName="Arial" x:Family="Swiss" ss:Bold="1"/>| && cl_
abap_char_utilities=>cr_lf &&
      |    <Interior ss:Color="#F7C7AC" ss:Pattern="Solid"/>| && cl_abap_ch
ar_utilities=>cr_lf &&
      |  </Style>| && cl_abap_char_utilities=>cr_lf &&

      |  <Style ss:ID="s71">| && cl_abap_char_utilities=>cr_lf &&
      |    <Alignment ss:Horizontal="Center" ss:Vertical="Top"/>| && cl_aba
p_char_utilities=>cr_lf &&
      |    <Borders>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Bottom" ss:LineStyle="Continuous" ss:Weigh
t="1"/>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Left" ss:LineStyle="Continuous" ss:Weight=
"1"/>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Right" ss:LineStyle="Continuous" ss:Weight
="1"/>| && cl_abap_char_utilities=>cr_lf &&
      |     <Border ss:Position="Top" ss:LineStyle="Continuous" ss:Weight="
1"/>| && cl_abap_char_utilities=>cr_lf &&
      |    </Borders>| && cl_abap_char_utilities=>cr_lf &&
      |  </Style>| && cl_abap_char_utilities=>cr_lf &&

      | </Styles>| && cl_abap_char_utilities=>cr_lf &&

* ===================== WORKSHEET =====================
      | <Worksheet ss:Name="Data">| && cl_abap_char_utilities=>cr_lf &&
      |  <Table ss:ExpandedColumnCount="26" ss:ExpandedRowCount="{ lv_exp_
rows }"| && cl_abap_char_utilities=>cr_lf &&
      |    x:FullColumns="1" x:FullRows="1" ss:StyleID="s62" ss:DefaultRowH
eight="15">| && cl_abap_char_utilities=>cr_lf &&

* ─── Column Widths ───
      |    <Column ss:StyleID="s62" ss:AutoFitWidth="0" ss:Width="40"/>| &&
 cl_abap_char_utilities=>cr_lf &&
      |    <Column ss:StyleID="s63" ss:AutoFitWidth="0" ss:Width="118"/>| &
& cl_abap_char_utilities=>cr_lf &&
      |    <Column ss:StyleID="s63" ss:Width="150"/>| && cl_abap_char_utili
ties=>cr_lf &&
      |    <Column ss:StyleID="s64" ss:AutoFitWidth="0" ss:Width="120"/>| &
& cl_abap_char_utilities=>cr_lf &&
      |    <Column ss:StyleID="s64" ss:Width="140"/>| && cl_abap_char_utili
ties=>cr_lf &&
      |    <Column ss:StyleID="s63" ss:AutoFitWidth="0" ss:Width="130"/>| &
```

```
& cl_abap_char_utilities=>cr_lf &&
      |   <Column ss:StyleID="s63" ss:AutoFitWidth="0" ss:Width="150"/>| &
& cl_abap_char_utilities=>cr_lf &&
      |   <Column ss:StyleID="s64" ss:AutoFitWidth="0" ss:Width="125"/>| &
& cl_abap_char_utilities=>cr_lf &&
      |   <Column ss:StyleID="s63" ss:AutoFitWidth="0" ss:Width="150"/>| &
& cl_abap_char_utilities=>cr_lf &&
      |   <Column ss:StyleID="s63" ss:AutoFitWidth="0" ss:Width="150"/>| &
& cl_abap_char_utilities=>cr_lf &&
      |   <Column ss:StyleID="s63" ss:Width="350"/>| && cl_abap_char_utili
ties=>cr_lf &&
      |   <Column ss:StyleID="s63" ss:AutoFitWidth="0" ss:Width="125" ss:S
pan="14"/>| && cl_abap_char_utilities=>cr_lf &&

* === Row 1: Header Row (Peach background, Bold) ===
      |   <Row ss:AutoFitHeight="0" ss:Height="40" ss:StyleID="s70">| && c
l_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">S No.</Data></Cell>| && cl_abap_ch
ar_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Plant</Data></Cell>| && cl_abap_ch
ar_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Type of Audit/Activity</Data></Cel
l>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Audit/Activity From</Data></Cell>|
 && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Audit Report received date</Data><
/Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Reference No</Data></Cell>| && cl_
abap_char_utilities=>cr_lf &&
      |    <Cell><ss:Data ss:Type="String" xmlns="http://www.w3.org/TR/REC
-html40"><B>Priority </B><Font>(Select from Dropdown)</Font></ss:Data></
Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Target Date</Data></Cell>| && cl_a
bap_char_utilities=>cr_lf &&
      |    <Cell><ss:Data ss:Type="String" xmlns="http://www.w3.org/TR/REC
-html40"><B>Location </B><Font>(Select from Dropdown)</Font></ss:Data></
Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Sub Location</Data></Cell>| && cl_
abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Observation Text</Data></Cell>| &&
 cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 1 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 2 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 3 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 4 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 5 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 6 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 7 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 8 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 9 (Use min 8 digits
)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
      |    <Cell><Data ss:Type="String">Implementor ID 10 (Use min 8 digit
s)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
```

```
333      |    <Cell><Data ss:Type="String">Implementor ID 11 (Use min 8 digit
     s)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
334      |    <Cell><Data ss:Type="String">Implementor ID 12 (Use min 8 digit
     s)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
335      |    <Cell><Data ss:Type="String">Implementor ID 13 (Use min 8 digit
     s)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
336      |    <Cell><Data ss:Type="String">Implementor ID 14 (Use min 8 digit
     s)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
337      |    <Cell><Data ss:Type="String">Implementor ID 15 (Use min 8 digit
     s)</Data></Cell>| && cl_abap_char_utilities=>cr_lf &&
338      |   </Row>| && cl_abap_char_utilities=>cr_lf &&
339
340  * === Data Rows (Row 2 onwards) ===
341      lv_data_rows &&
342
343      |  </Table>| && cl_abap_char_utilities=>cr_lf &&
344
345  * ─── Worksheet Options (Freeze Row 1 header) ───
346      |  <WorksheetOptions xmlns="urn:schemas-microsoft-com:office:excel">
     | && cl_abap_char_utilities=>cr_lf &&
347      |   <Print>| && cl_abap_char_utilities=>cr_lf &&
348      |    <ValidPrinterInfo/>| && cl_abap_char_utilities=>cr_lf &&
349      |    <PaperSizeIndex>9</PaperSizeIndex>| && cl_abap_char_utilities=>
     cr_lf &&
350      |    <HorizontalResolution>600</HorizontalResolution>| && cl_abap_ch
     ar_utilities=>cr_lf &&
351      |    <VerticalResolution>600</VerticalResolution>| && cl_abap_char_u
     tilities=>cr_lf &&
352      |   </Print>| && cl_abap_char_utilities=>cr_lf &&
353      |   <Selected/>| && cl_abap_char_utilities=>cr_lf &&
354      |   <FreezePanes/>| && cl_abap_char_utilities=>cr_lf &&
355      |   <FrozenNoSplit/>| && cl_abap_char_utilities=>cr_lf &&
356      |   <SplitHorizontal>1</SplitHorizontal>| && cl_abap_char_utilities=
     >cr_lf &&
357      |   <TopRowBottomPane>1</TopRowBottomPane>| && cl_abap_char_utilitie
     s=>cr_lf &&
358      |   <Panes>| && cl_abap_char_utilities=>cr_lf &&
359      |    <Pane>| && cl_abap_char_utilities=>cr_lf &&
360      |     <Number>3</Number>| && cl_abap_char_utilities=>cr_lf &&
361      |     <ActiveRow>1</ActiveRow>| && cl_abap_char_utilities=>cr_lf &&
362      |     <ActiveCol>6</ActiveCol>| && cl_abap_char_utilities=>cr_lf &&
363      |    </Pane>| && cl_abap_char_utilities=>cr_lf &&
364      |   </Panes>| && cl_abap_char_utilities=>cr_lf &&
365      |   <ProtectObjects>False</ProtectObjects>| && cl_abap_char_utilitie
     s=>cr_lf &&
366      |   <ProtectScenarios>False</ProtectScenarios>| && cl_abap_char_util
     ities=>cr_lf &&
367      |  </WorksheetOptions>| && cl_abap_char_utilities=>cr_lf &&
368
369  * === DATA VALIDATIONS (DROPDOWNS) ===
370
371  * ─── Priority Dropdown: Column G (Col 7) ───
372      |  <DataValidation xmlns="urn:schemas-microsoft-com:office:excel">|
     && cl_abap_char_utilities=>cr_lf &&
373      |   <Range>{ lv_row_range }</Range>| && cl_abap_char_utilities=>cr_l
     f &&
374      |   <Type>List</Type>| && cl_abap_char_utilities=>cr_lf &&
375      |   <CellRangeList/>| && cl_abap_char_utilities=>cr_lf &&
376      |   <Value>| && lv_priority_val && |</Value>| && cl_abap_char_utilit
     ies=>cr_lf &&
377      |  </DataValidation>| && cl_abap_char_utilities=>cr_lf &&
```

```
* ──┐ Location Dropdown: Column I (Col 9) ───
     | <DataValidation xmlns="urn:schemas-microsoft-com:office:excel">| && cl_abap_char_utilities=>cr_lf &&
     |   <Range>{ lv_loc_range }</Range>| && cl_abap_char_utilities=>cr_lf &&
     |   <Type>List</Type>| && cl_abap_char_utilities=>cr_lf &&
     |   <CellRangeList/>| && cl_abap_char_utilities=>cr_lf &&
     |   <Value>| && lv_location_val && |</Value>| && cl_abap_char_utilities=>cr_lf &&
     | </DataValidation>| && cl_abap_char_utilities=>cr_lf &&

     | </Worksheet>| && cl_abap_char_utilities=>cr_lf &&
     |</Workbook>|.

ENDFORM.



*&---------------------------------------------------------------------*
*& Form DISPLAY_ALV
*&---------------------------------------------------------------------*
FORM display_alv USING iv_count TYPE i.

  DATA: lo_cols    TYPE REF TO cl_salv_columns_table,
        lo_col     TYPE REF TO cl_salv_column,
        lo_funcs   TYPE REF TO cl_salv_functions_list,
        lo_disp    TYPE REF TO cl_salv_display_settings,
        lo_events  TYPE REF TO cl_salv_events_table,
        lo_handler TYPE REF TO lcl_event_handler.

  TRY.
      cl_salv_table=>factory(
        IMPORTING r_salv_table = go_alv
        CHANGING  t_table      = gt_upload ).

*     Enable toolbar
      lo_funcs = go_alv->get_functions( ).
      lo_funcs->set_all( abap_true ).

*     ──────────────────────────────────────────────────────────────
*     Add custom buttons: SAVE AS DRAFT and SAVE
*     ──────────────────────────────────────────────────────────────
      TRY.
*         Save as Draft button (icon: floppy disk with pencil)
          lo_funcs->add_function(
            name     = 'SAVE_DRAFT'
            icon     = '@4G@'           " ICON_SYSTEM_SAVE
            text     = 'Save as Draft'
            tooltip  = 'Save data as Draft'
            position = if_salv_c_function_position=>right_of_salv_functions ).

*         Save button (icon: green check/save)
          lo_funcs->add_function(
            name     = 'SAVE_FINAL'
            icon     = '@2L@'           " ICON_OKAY
            text     = 'Save'
            tooltip  = 'Save data (Final)'
            position = if_salv_c_function_position=>right_of_salv_functions ).
```

```
434
435          CATCH cx_salv_wrong_call cx_salv_existing.
436        ENDTRY.
437
438  *     ____________________________________________________________
439  *     Register event handler for custom buttons
440  *     ____________________________________________________________
441        lo_events = go_alv->get_event( ).
442        CREATE OBJECT lo_handler.
443        SET HANDLER lo_handler->on_user_command FOR lo_events.
444
445  *     Optimize column widths
446        lo_cols = go_alv->get_columns( ).
447        lo_cols->set_optimize( abap_true ).
448
449  *     Set column headers
450        TRY.
451            lo_col = lo_cols->get_column( 'SNO' ).
452            lo_col->set_short_text( 'S No.' ).
453            lo_col->set_medium_text( 'S No.' ).
454            lo_col->set_long_text( 'S No.' ).
455
456            lo_col = lo_cols->get_column( 'PLANT' ).
457            lo_col->set_short_text( 'Plant' ).
458            lo_col->set_medium_text( 'Plant' ).
459            lo_col->set_long_text( 'Plant' ).
460
461            lo_col = lo_cols->get_column( 'AUTYPE' ).
462            lo_col->set_short_text( 'AuditType' ).
463            lo_col->set_medium_text( 'Audit Type' ).
464            lo_col->set_long_text( 'Type of Audit/Activity' ).
465
466            lo_col = lo_cols->get_column( 'AUFROM' ).
467            lo_col->set_short_text( 'From' ).
468            lo_col->set_medium_text( 'Audit From' ).
469            lo_col->set_long_text( 'Audit/Activity From' ).
470
471            lo_col = lo_cols->get_column( 'RCDATE' ).
472            lo_col->set_short_text( 'Recv Date' ).
473            lo_col->set_medium_text( 'Received Date' ).
474            lo_col->set_long_text( 'Audit Report Received Date' ).
475
476            lo_col = lo_cols->get_column( 'REFNO' ).
477            lo_col->set_short_text( 'Ref No' ).
478            lo_col->set_medium_text( 'Reference No' ).
479            lo_col->set_long_text( 'Reference No' ).
480
481            lo_col = lo_cols->get_column( 'PRIORITY' ).
482            lo_col->set_short_text( 'Priority' ).
483            lo_col->set_medium_text( 'Priority' ).
484            lo_col->set_long_text( 'Priority' ).
485
486            lo_col = lo_cols->get_column( 'TGTDATE' ).
487            lo_col->set_short_text( 'Tgt Date' ).
488            lo_col->set_medium_text( 'Target Date' ).
489            lo_col->set_long_text( 'Target Date' ).
490
491            lo_col = lo_cols->get_column( 'LOCATION' ).
492            lo_col->set_short_text( 'Location' ).
493            lo_col->set_medium_text( 'Location' ).
494            lo_col->set_long_text( 'Location' ).
```

```

          lo_col = lo_cols->get_column( 'SUBLOC' ).
          lo_col->set_short_text( 'Sub Loc' ).
          lo_col->set_medium_text( 'Sub Location' ).
          lo_col->set_long_text( 'Sub Location' ).

          lo_col = lo_cols->get_column( 'OBSTEXT' ).
          lo_col->set_short_text( 'Obs Text' ).
          lo_col->set_medium_text( 'Observation' ).
          lo_col->set_long_text( 'Observation Text' ).

        CATCH cx_salv_not_found.
      ENDTRY.

*     Display settings
      lo_disp = go_alv->get_display_settings( ).
      lo_disp->set_striped_pattern( abap_true ).
      lo_disp->set_list_header( |EHSM Audit Upload - { iv_count } Records| ).

*     Display ALV
      go_alv->display( ).

    CATCH cx_salv_msg INTO DATA(lx_msg).
      MESSAGE lx_msg->get_text( ) TYPE 'E'.
  ENDTRY.

ENDFORM.

*&---------------------------------------------------------------------*
*& Form SAVE_AS_DRAFT
*& Called when user clicks "Save as Draft" button
*&---------------------------------------------------------------------*
FORM save_as_draft.

  DATA: lv_count TYPE i.

  lv_count = lines( gt_upload ).

  IF lv_count = 0.
    MESSAGE 'No data to save' TYPE 'S' DISPLAY LIKE 'E'.
    RETURN.
  ENDIF.

* Confirmation popup
  DATA: lv_answer TYPE c.
  CALL FUNCTION 'POPUP_TO_CONFIRM'
    EXPORTING
      titlebar              = 'Save as Draft'
      text_question         = |Save { lv_count } records as Draft?|
      text_button_1         = 'Yes'
      icon_button_1         = 'ICON_OKAY'
      text_button_2         = 'No'
      icon_button_2         = 'ICON_CANCEL'
      display_cancel_button = ' '
    IMPORTING
      answer                = lv_answer.

  IF lv_answer <> '1'.
    RETURN.
  ENDIF.
```

```

* ────────────────────────────────────────────────────────────
* >>> ADD YOUR DRAFT SAVE LOGIC HERE <
* Example: Insert into Z-table with status = 'DRAFT'
* ────────────────────────────────────────────────────────────
* LOOP AT gt_upload INTO DATA(ls_data).
*   " Populate Z-table structure
*   " ls_ztable-status = 'DRAFT'.
*   " INSERT yehs_observations FROM ls_ztable.
* ENDLOOP.
* COMMIT WORK AND WAIT.
* ────────────────────────────────────────────────────────────

  MESSAGE |{ lv_count } records saved as Draft successfully| TYPE 'S'.

ENDFORM.

*&──────────────────────────────────────────────────────────────────────*
*& Form SAVE_FINAL
*& Called when user clicks "Save" button
*&──────────────────────────────────────────────────────────────────────*
FORM save_final.

  DATA: lv_count TYPE i.

  lv_count = lines( gt_upload ).

  IF lv_count = 0.
    MESSAGE 'No data to save' TYPE 'S' DISPLAY LIKE 'E'.
    RETURN.
  ENDIF.

* Confirmation popup
  DATA: lv_answer TYPE c.
  CALL FUNCTION 'POPUP_TO_CONFIRM'
    EXPORTING
      titlebar              = 'Save (Final)'
      text_question         = |Save { lv_count } records as Final? This cannot be undone.|
      text_button_1         = 'Yes'
      icon_button_1         = 'ICON_OKAY'
      text_button_2         = 'No'
      icon_button_2         = 'ICON_CANCEL'
      display_cancel_button = ' '
    IMPORTING
      answer                = lv_answer.

  IF lv_answer <> '1'.
    RETURN.
  ENDIF.

* ────────────────────────────────────────────────────────────
* >>> ADD YOUR FINAL SAVE LOGIC HERE <
* Example: Insert into Z-table with status = 'FINAL'
*          or call BAPI to create audit observations
* ────────────────────────────────────────────────────────────
* LOOP AT gt_upload INTO DATA(ls_data).
*   " Populate Z-table structure
*   " ls_ztable-status = 'FINAL'.
*   " INSERT yehs_observations FROM ls_ztable.
* ENDLOOP.
```

```
* COMMIT WORK AND WAIT.
* ---------------------------------------------------------------------

  MESSAGE |{ lv_count } records saved (Final) successfully| TYPE 'S'.

ENDFORM.
```

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